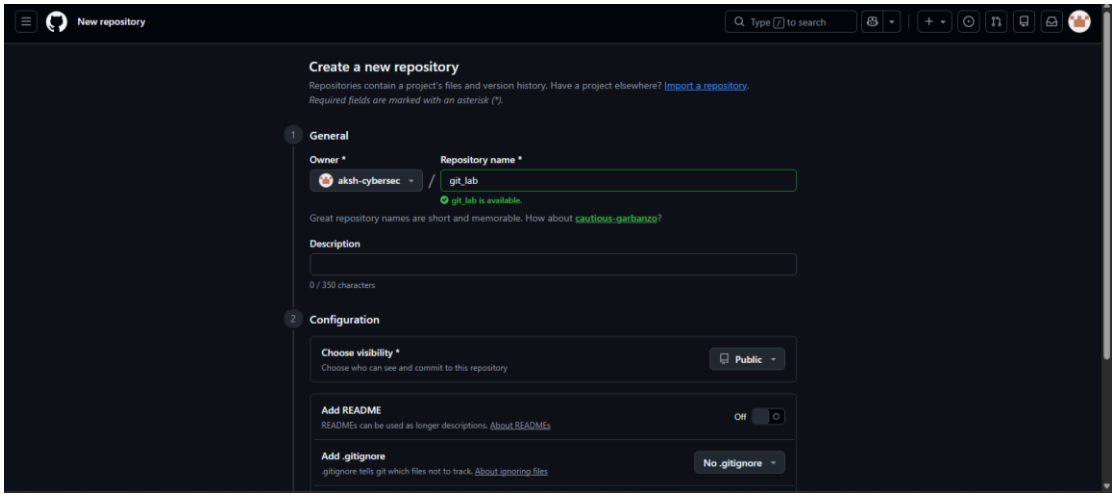
Assignment 1: (Git & GitHub)

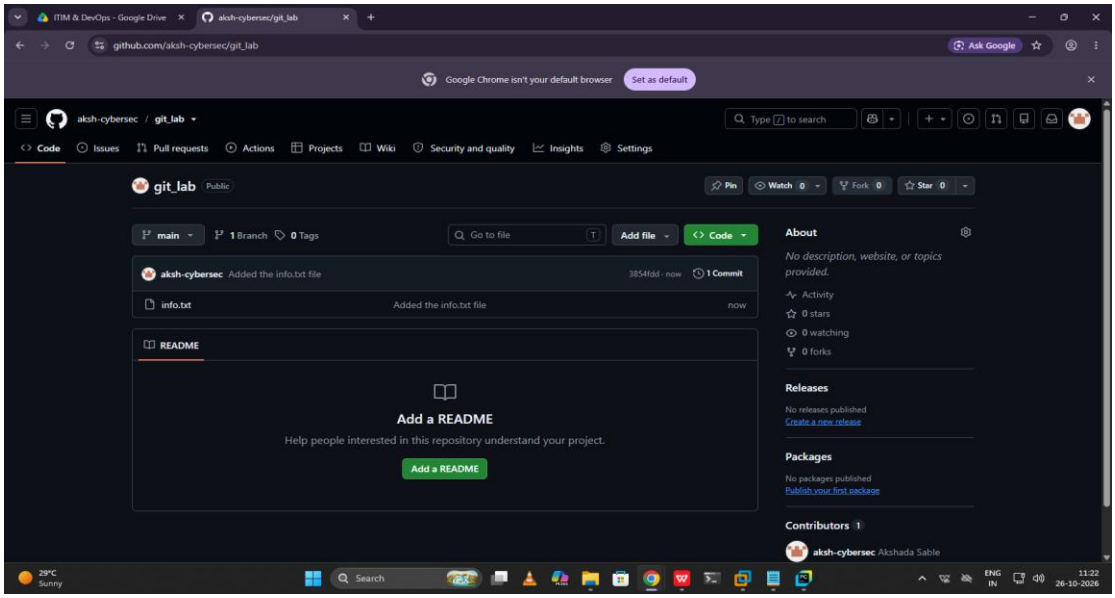
Q1. Create and Initialize Git Repository Create a new repository named git_lab on your local system. Initialize it using Git. Create a file info.txt and add your name and date. Check the repository status. Add the file to staging and commit with message "Initial commit" .



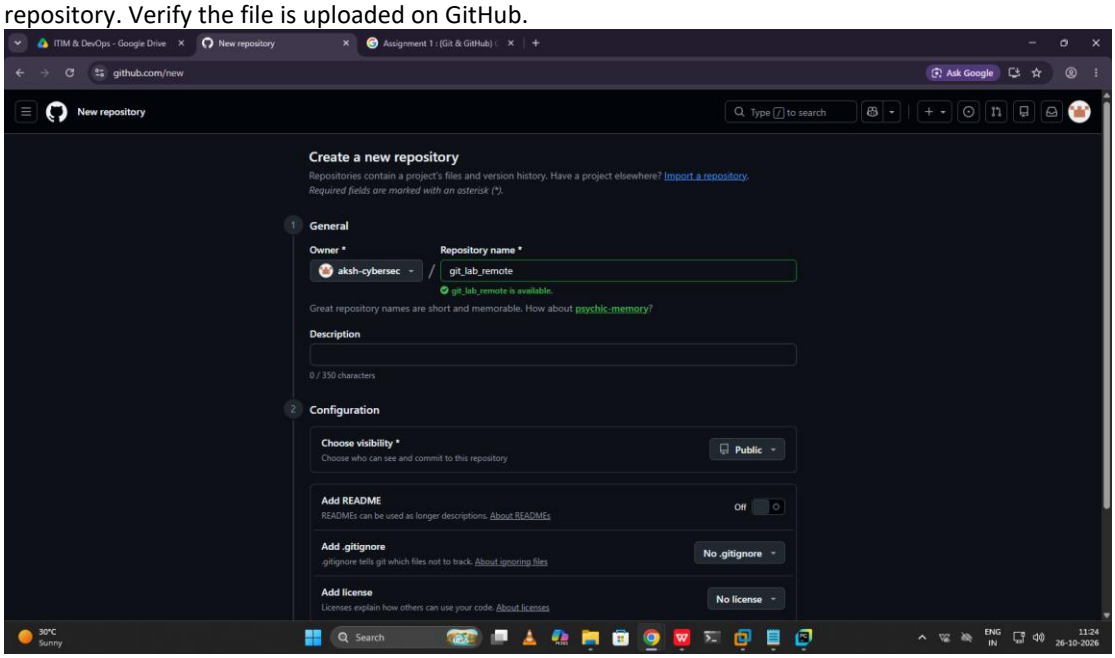

```
root@server-virtual-machine:~# mkdir Assignment1
root@server-virtual-machine:~# cd Assignment1
root@server-virtual-machine:~/Assignment1# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /root/Assignment1/.git/
root@server-virtual-machine:~/Assignment1# nano info.txt
```

```
root@server-virtual-machine:~/Assignment1# git add .
root@server-virtual-machine:~/Assignment1# git commit -m "Added the info.txt file"
[master (root-commit) 3854fdd] Added the info.txt file
 1 file changed, 2 insertions(+)
 create mode 100644 info.txt
root@server-virtual-machine:~/Assignment1# git branch -M  main
root@server-virtual-machine:~/Assignment1# git remote add origin https://github.com/aksh-cybersec/git_lab.git
root@server-virtual-machine:~/Assignment1# git push -u origin main
Username for 'https://github.com': aksh-cybersec
Password for 'https://aksh-cybersec@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 267 bytes | 267.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/aksh-cybersec/git_lab.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
root@server-virtual-machine:~/Assignment1#
```

Assignment 1: (Git & GitHub)

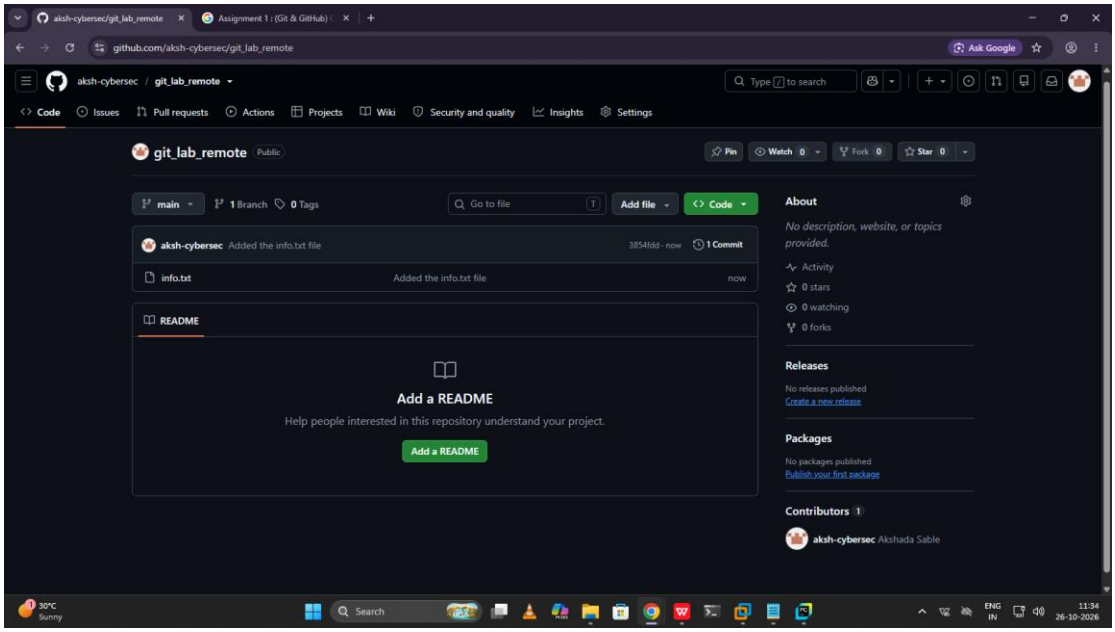


Q2. Connect to GitHub and Push Code Create a repository named git_lab_remote on GitHub. Link your local repository (git_lab) to the remote repository. Push your committed file to the remote repository. Verify the file is uploaded on GitHub.

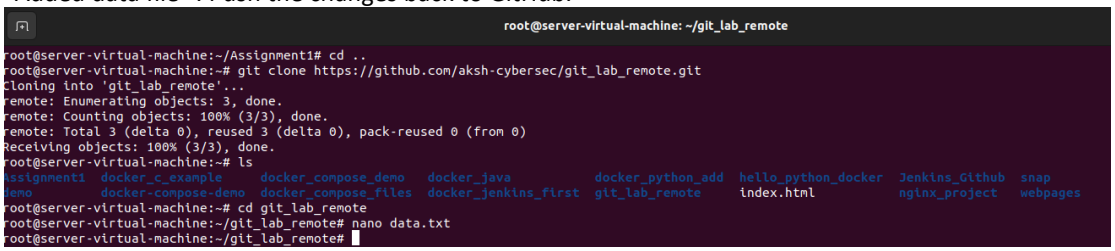


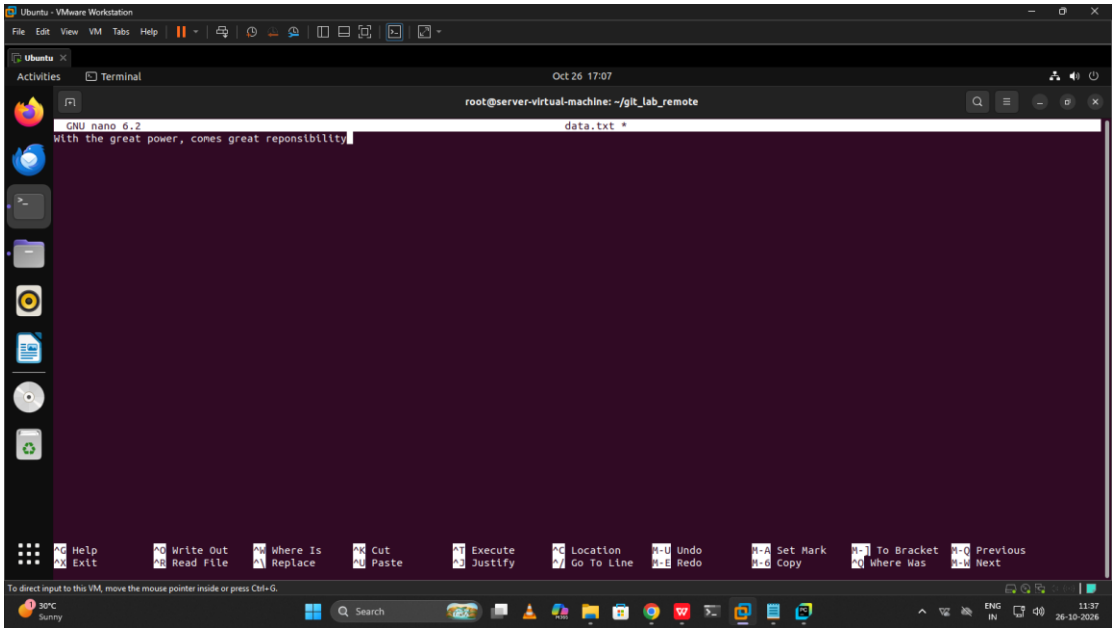

```
root@server-virtual-machine:~/Assignment1# git remote remove origin
root@server-virtual-machine:~/Assignment1# git remote add origin https://github.com/aksh-cybersec/git_lab_remote.git
root@server-virtual-machine:~/Assignment1# git remote -v
origin  https://github.com/aksh-cybersec/git_lab_remote.git (fetch)
origin  https://github.com/aksh-cybersec/git_lab_remote.git (push)
root@server-virtual-machine:~/Assignment1# git push -u origin main
Username for 'https://github.com': aksh-cybersec
Password for 'https://aksh-cybersec@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 267 bytes | 267.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/aksh-cybersec/git_lab_remote.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
root@server-virtual-machine:~/Assignment1#
```

Assignment 1: (Git & GitHub)



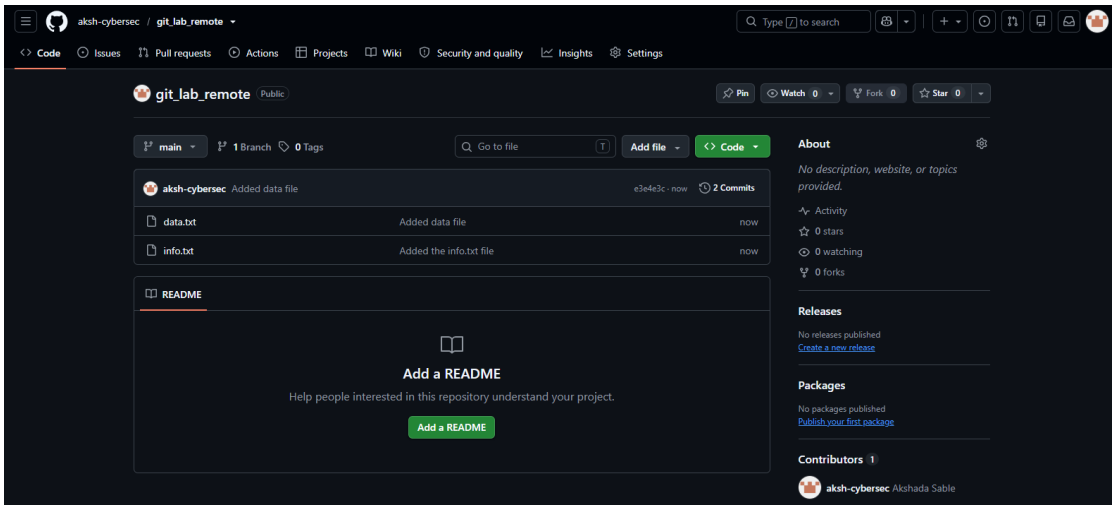
Q3. Clone Repository and Make Changes Clone the repository git_lab_remote to a new directory. Create a new file data.txt and add some sample content. Add and commit the file with message "Added data file" . Push the changes back to GitHub.





Assignment 1: (Git & GitHub)

```
root@server-virtual-machine:~/git_lab_remote# git add data.txt
root@server-virtual-machine:~/git_lab_remote# git commit -m "Added data file"
[main e3e4e3c] Added data file
 1 file changed, 1 insertion(+)
 create mode 100644 data.txt
root@server-virtual-machine:~/git_lab_remote# git branch -M main
root@server-virtual-machine:~/git_lab_remote# git remote add origin https://github.com/aksh-cybersec/git_lab_remote.git
error: remote origin already exists.
root@server-virtual-machine:~/git_lab_remote# git push -u origin main
Username for 'https://github.com': aksh-cybersec
Password for 'https://aksh-cybersec@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 323 bytes | 323.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/aksh-cybersec/git_lab_remote.git
   3854fdd..e3e4e3c  main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
root@server-virtual-machine:~/git_lab_remote#
```



Q4. Branching and Merging Create a new branch named feature . Switch to the feature branch. Modify info.txt (add new line: “Feature branch update”). Commit the changes. Switch back to main branch. Merge the feature branch into main . Delete the feature branch after merge.

```
root@server-virtual-machine:~/git_lab_remote# git checkout -b feature
Switched to a new branch 'feature'
root@server-virtual-machine:~/git_lab_remote# echo "Feature branch update" >> info.txt
root@server-virtual-machine:~/git_lab_remote# git add info.txt
root@server-virtual-machine:~/git_lab_remote# git commit -m "Feature branch update"
[feature e37fb42] Feature branch update
 1 file changed, 1 insertion(+)
root@server-virtual-machine:~/git_lab_remote# git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
root@server-virtual-machine:~/git_lab_remote# git merge feature
Updating e3e4e3c..e37fb42
Fast-forward
 info.txt | 1 +
 1 file changed, 1 insertion(+)
root@server-virtual-machine:~/git_lab_remote# git branch -d feature
Deleted branch feature (was e37fb42).
root@server-virtual-machine:~/git_lab_remote#
```

